



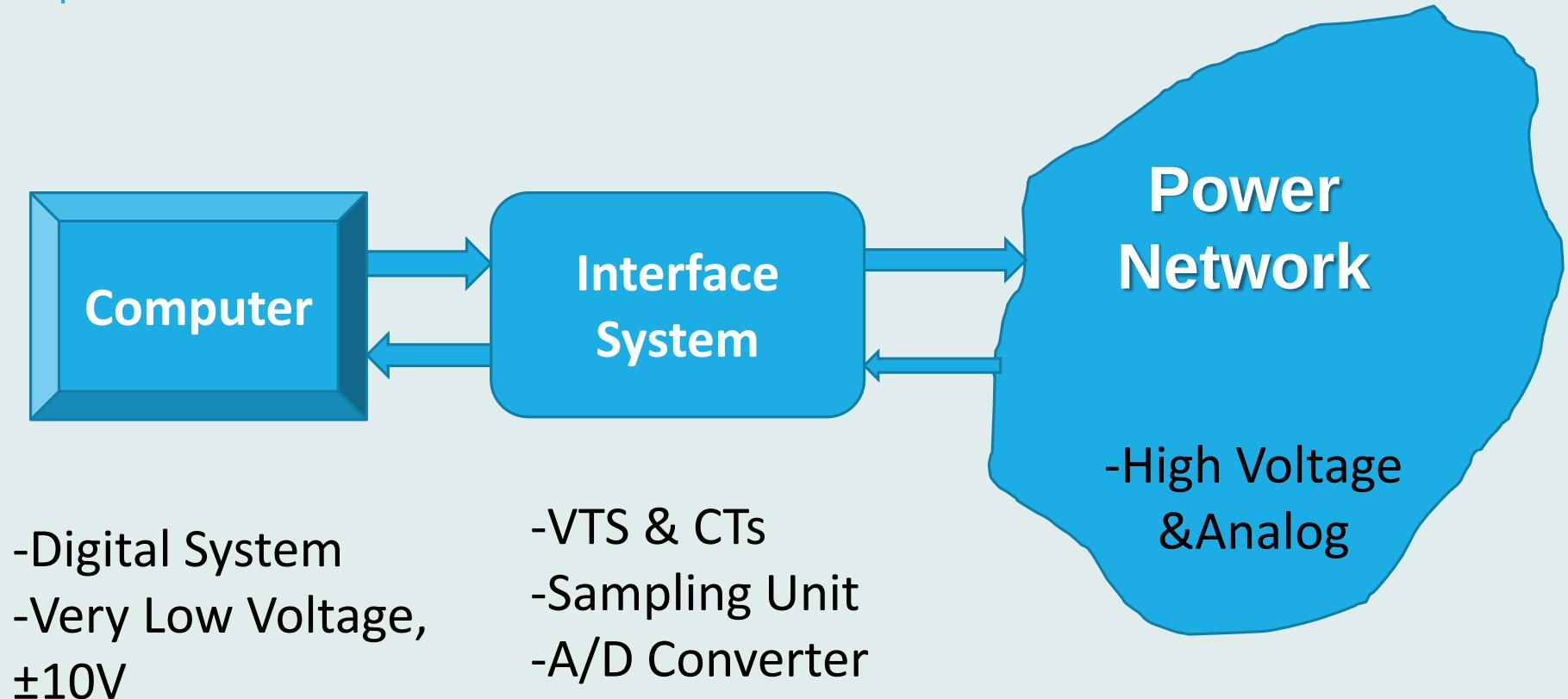
ELEK455E COMPUTER NETWORKS FOR POWER SYSTEMS

**ELK455E ELEKTRİK ŞEBEKELERİ İÇİN
BİLGİSAYAR AĞLARI**

CHAPTER 0: AN INTRODUCTION TO COMPUTER SYSTEMS

Computer Applications in Power Systems

Computers are used in power systems for off-line, on-line and real-time applications.



Computers Applications in Power Systems (Real –Time)

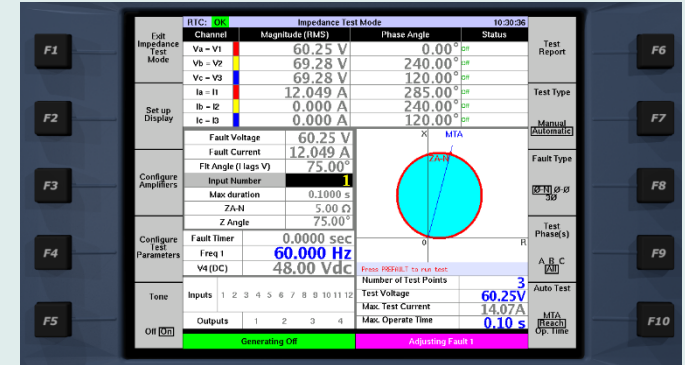
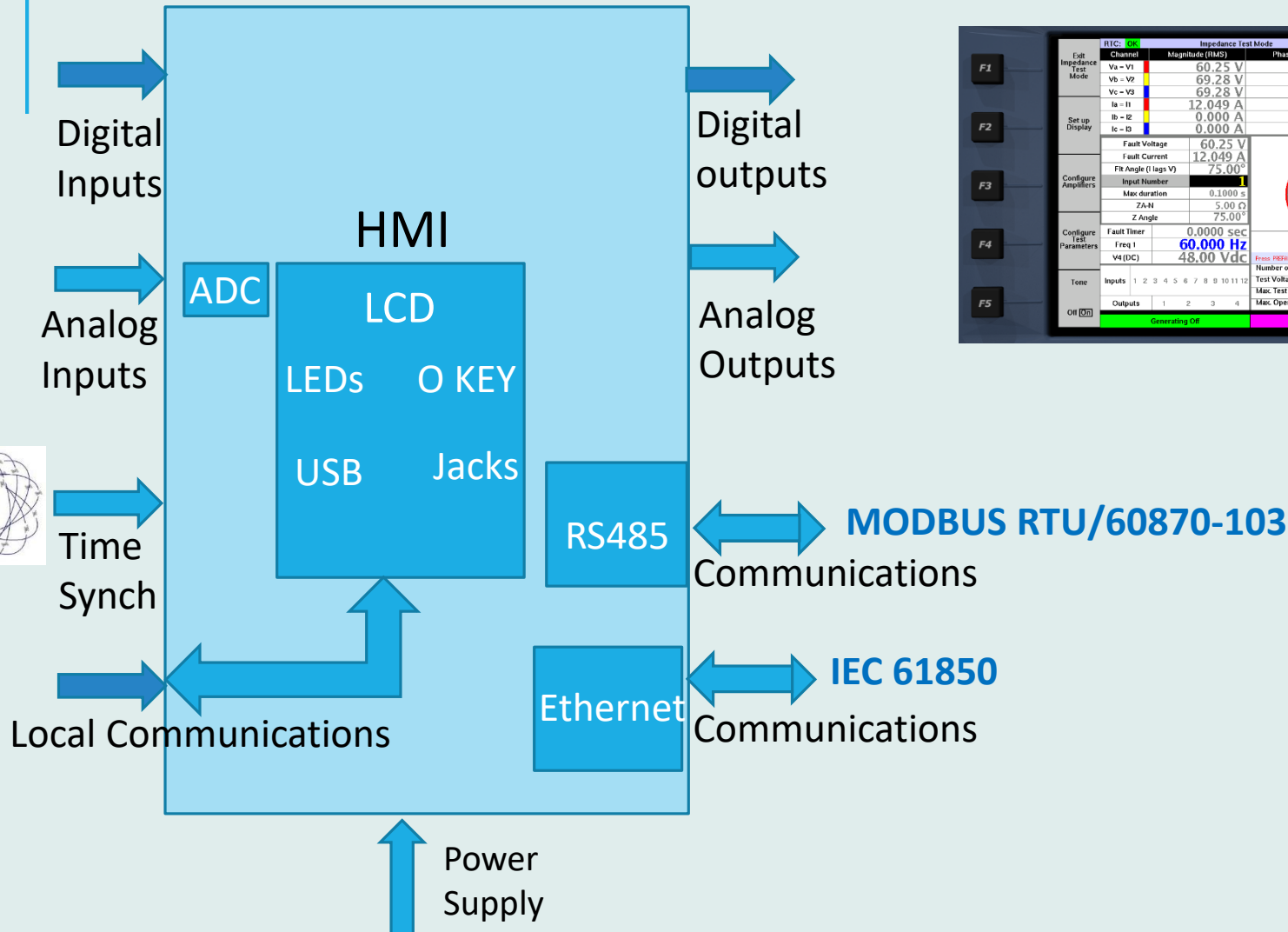
A Computer designed for power system application is known as a Computer relay (a relay Computer), or a Protective Relay or an Intelligent Electronic Device (IED).

And used for:

- Power System Measurement / Metering,
- Power System Monitoring,
- Power System Protection,
- Power System Control,
- Power System Communications,
- and Power System Management.

Relay Computer:

The layout of Intelligent Electronic Devices



COMPUTER SYSTEMS



The modern computer is high-speed electronic computational machine that accepts information in the form of data and instructions through some input device and processes this information with arithmetic and logic operation of a program stored in its memory.

The results of processing can be displayed, stored or recovered using suitable output devices or transmitted to another location.

COMPUTER SYSTEMS

Hardware: the physical components of the computer, including the computer itself and Peripheral I/O devices.

- Central processing unit (CPU or Microprocessor Unit),
- Main memory (RAM, and ROM),
- Input/output (I/O) unit.

The three components – MPU, memory, and I/O – are connected by a group of wires called the BUS

- Address bus
- Control bus
- Data bus

These are used to provide data flow among three units.

COMPUTER SYSTEMS

Software:

1.**System software**: the programs that run the computer such as: OS, compilers, drivers.

2.**Application software** required for the different applications such as: Word, MatLab, Media player, Latex ...

Data are facts, or unorganized raw materials, which can be made up of words, numbers, images, or sounds. For instance: Sampled values of the voltage and current signals.

Information: data that has been processed. For example, calculated phaser values of voltage and current signals.

COMPUTER SYSTEMS/ BINARY SYSTEM

Bit (digit): the smallest unit of data that a computer uses (microprocessors). It can be used to represent two states of information, such as Yes or No (0,1), open or close (0,1).

Byte: is equal to 8 Bits. A Byte can represent 256 (2^8) states of information, for example, numbers or a combination of numbers and letters. 1 Byte could be equal to one character.

Word: the number of bits that a CPU can process at one time depends on the width of the CPU's registers and that of the data bus.

- if the width of the data bus is 16 bits, then a word is 16 bits; -
- if the width of the data bus is 32 bits, then a word is 32 bits.

- Usually 16-bits is called a **word**. 32-bits is called a **double word**, 64-bits is called a **long word**.

BINARY SYSTEM

1 byte = 8 bits

1 kilobyte (K / Kb) = 1,024 bytes

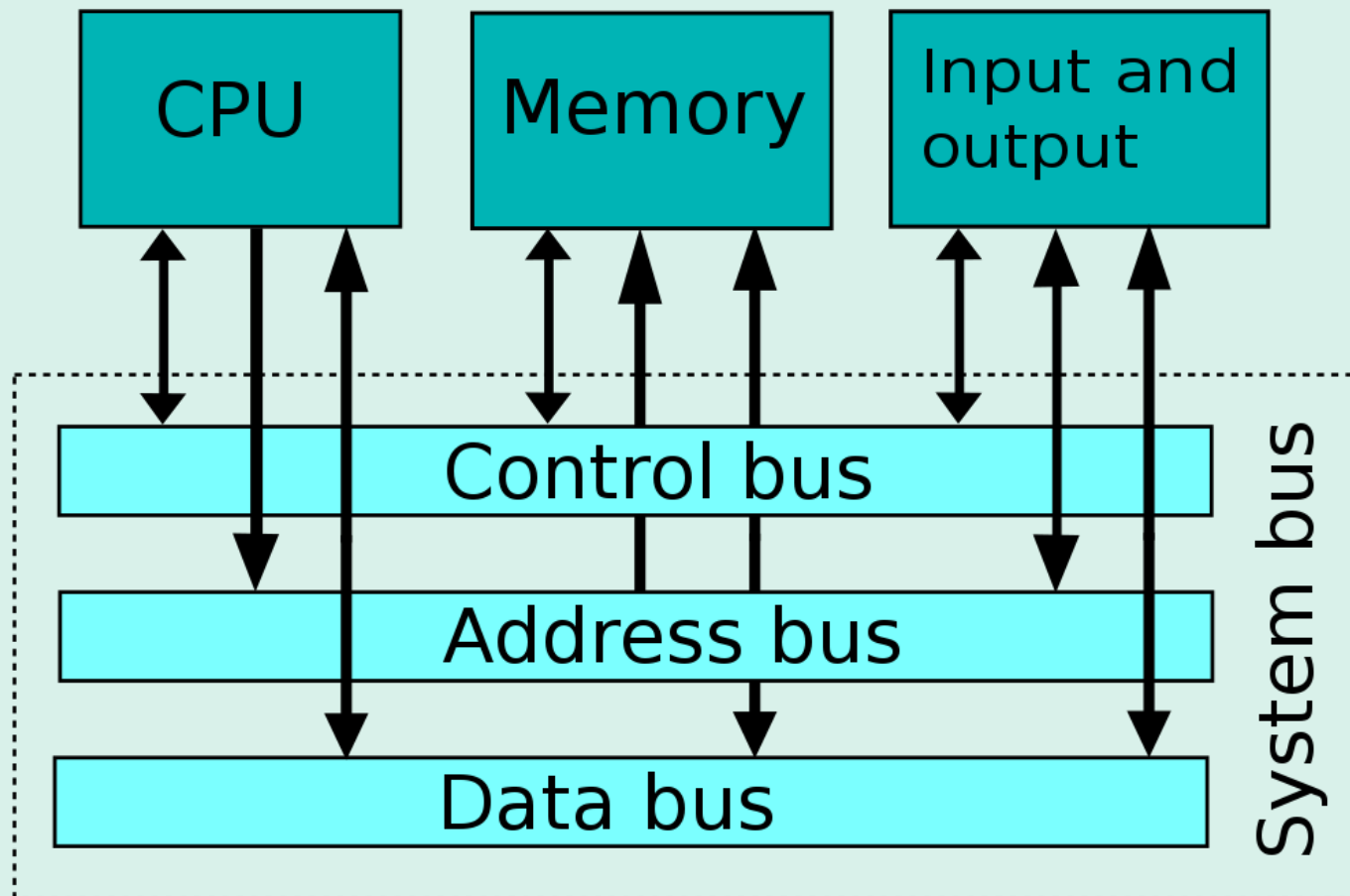
1 megabyte (M / MB) = 1,048,576 bytes

1 gigabyte (G / GB) = 1,073,741,824 bytes

1 terabyte (T / TB) = 1,099,511,627,776 bytes

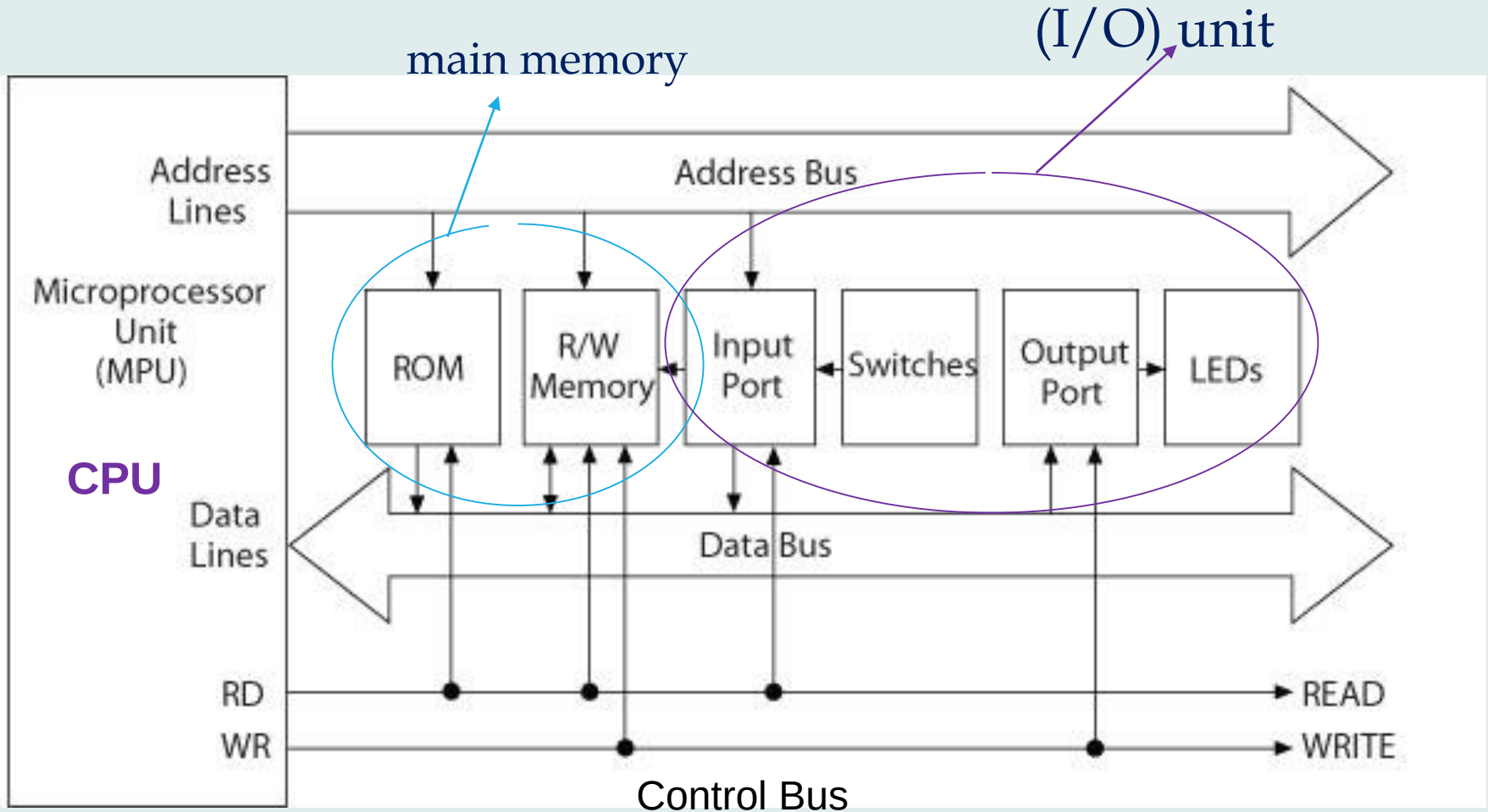
COMPUTER-BASED SYSTEM

A Basic computer system consist of a Central processing unit (CPU or Microprocessor), main memory (RAM and ROM), input/output (I/O) unit.



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BASIC COMPONENTS OF A MICROCOMPUTER SYSTEM

1. CPU - Central Processing Unit/Microprocessor

- the portion of a computer system that carries out the instructions of a computer program.
- the primary element carrying out the computer's functions. It is the unit that reads and executes program instructions.
- The data in the instruction tells the processor what to do.



CPU/MPU MICROPROCESSORS

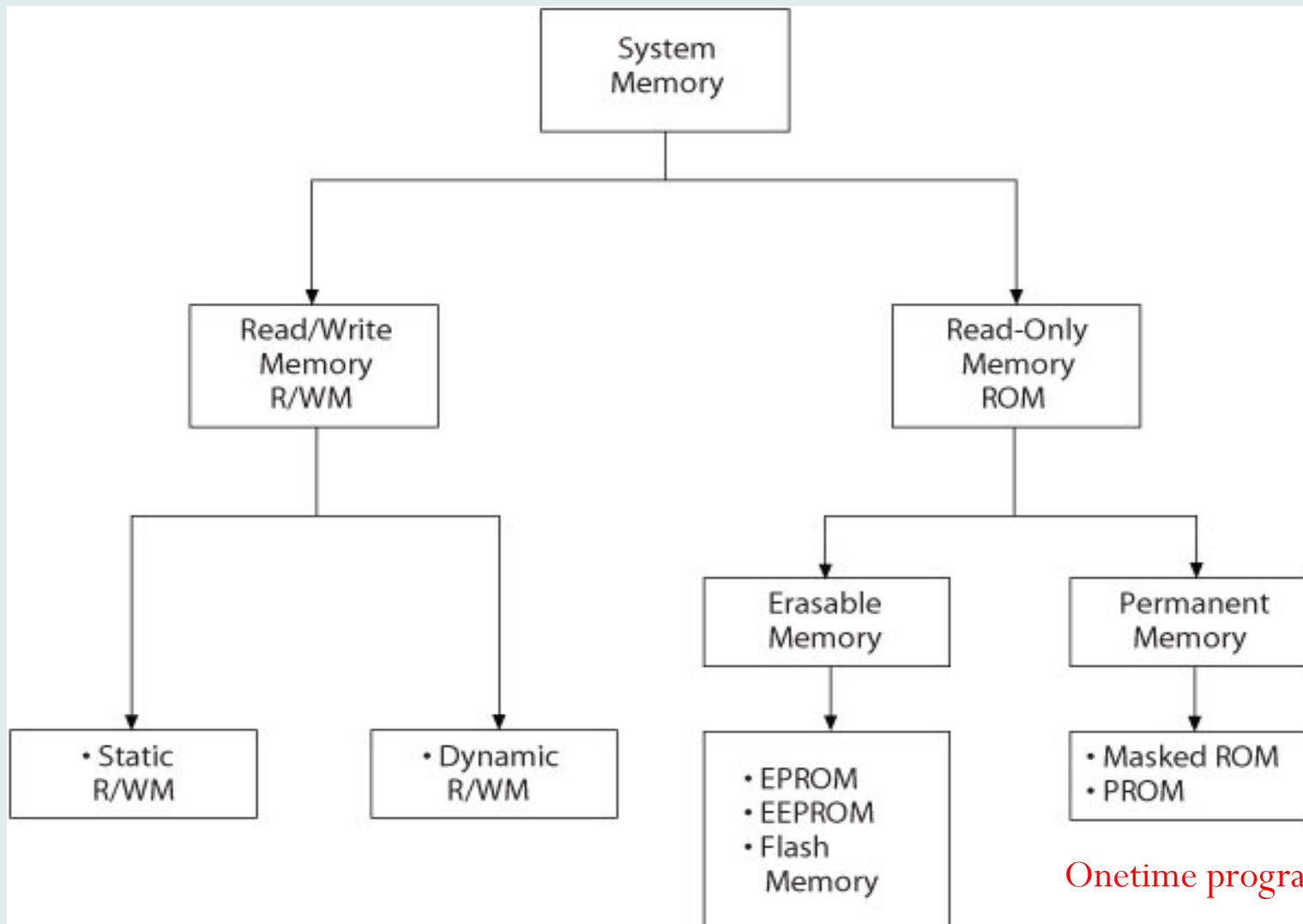
- The world's first microprocessor, the Intel 4004, was a 4-bit microprocessor—a programmable controller on a chip—that was meager by today's standards.
- It addressed a mere 4096 4-bit memory locations.
- Its instruction set contained only 45 different instructions.
- Microprocessors that are common today include the 8086/8088, which were the first 16-bit microprocessors.
- Following these early 16-bit machines were the 80286, 80386, 80486, Pentium, Pentium Pro, Pentium II, Pentium III, Pentium 4, and Core i3- i5, i7, i9 processors.



2. Main Memory

- Physical devices used to store data or programs (sequences of instructions) on a temporary or permanent basis for use in an electronic digital computer.
- Computer main memory comes in two principal varieties: random-access memory (RAM) and read-only memory (ROM).
- RAM can be read and written to anytime the CPU commands it, but ROM is pre-loaded with data and software that never changes, so the CPU can only read from it.
- ROM is typically used to store the computer's initial start-up instructions.
- In general, the contents of RAM are erased when the power to the computer is turned off, but ROM retains its data indefinitely.
- The ROM contains a specialized program called the BIOS that orchestrates loading the computer's OS from the hard disk drive into RAM whenever the computer is turned on or reset.

Memory Classification



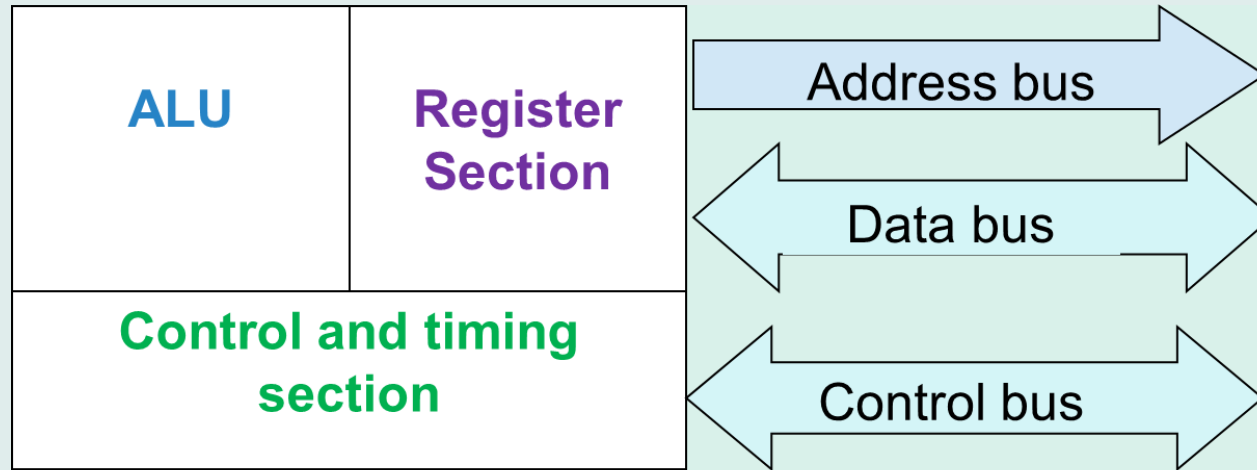
Onetime programmable

Electrically Erasable
PROM

3. I/O Unit

- **Input/output (I/O)**, refers to the communication between a computer, and the outside world possibly a human, or another information processing system.
- Inputs are the signals or data received by the system, and outputs are the signals or data sent from it.
- Devices that provide input or output to the computer are called peripherals
- On a typical personal computer, peripherals include input devices like the keyboard and mouse, and output devices such as the display and printer. Hard disk drives, floppy disk drives and optical disc drives serve as both input and output devices. Computer networking is another form of I/O.

Internal Structure Of Microprocessor



The register section/array consists completely of circuitry used to **temporarily store data or program codes until they** are sent to the ALU or to the control section or to memory.

The number of registers are different for any particular CPU and the more register a CPU have will result in easier programming tasks.

Registers are normally measured by the number of bits they can hold, for example, an "8-bit register" or a "32-bit register".

Aritmetic And Logic Unit (ALU)

The component that performs the arithmetic and logical operations.

the most important components in a microprocessor, and is typically the part of the processor that is designed first.

able to perform the basic logical operations (AND, OR), including the addition operation.

The inclusion of inverters on the inputs enables the same ALU hardware to perform the subtraction operation (adding an inverted operand), and the operations NAND and NOR.

CONTROL UNIT

The circuitry that controls the flow of information through the processor, and coordinates the activities of the other units within it.

In a way, it is the "brain within the brain", as it controls what happens inside the processor, which in turn controls the rest of the PC.

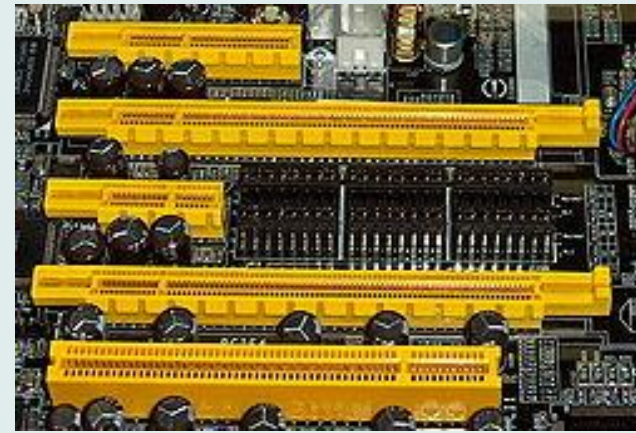
On a regular processor, the control unit performs the tasks of fetching, decoding, managing execution and then storing results.

BUS SYSTEM

a subsystem that transfers data between computer components inside a computer or between computers.

- The three components – MPU, memory, and I/O – are connected by a group of wires called the BUS
- **Address bus**
 - consists of 16, 20, 24, or 32 parallel signal lines (wires) - unidirectional
 - these lines contain the address of the memory location to read or written
- **Control bus**
 - consists of 4 to 10 (or more) parallel signal lines
 - CPU sends signals along these lines to memory and to I/O ports
 - examples: Memory Read, Memory Write, I/O Read, I/O Write
- **Data bus**
 - consists of 8,16, or 32 parallel signal lines
 - bi-directional
 - only one device at a time can have its outputs enabled,
 - this requires the devices to have three-state output

A bus is a communication pathway connecting two or more devices



The data bus is 'bi-directional' data or instruction codes from memory or input/output are transferred into the microprocessor the result of an operation or computation is sent out from the microprocessor to the memory or input/output

Depending on the particular microprocessor, the data bus can handle 8 bit, 16 bit, 32 bit or 64 bit data.

The address bus is 'unidirectional', over which the microprocessor sends an address code to the memory or input/output.

The size (width) of the address bus is specified by the number of bits it can handle.

The more bits there are in the address bus, the more memory locations a microprocessor can access.

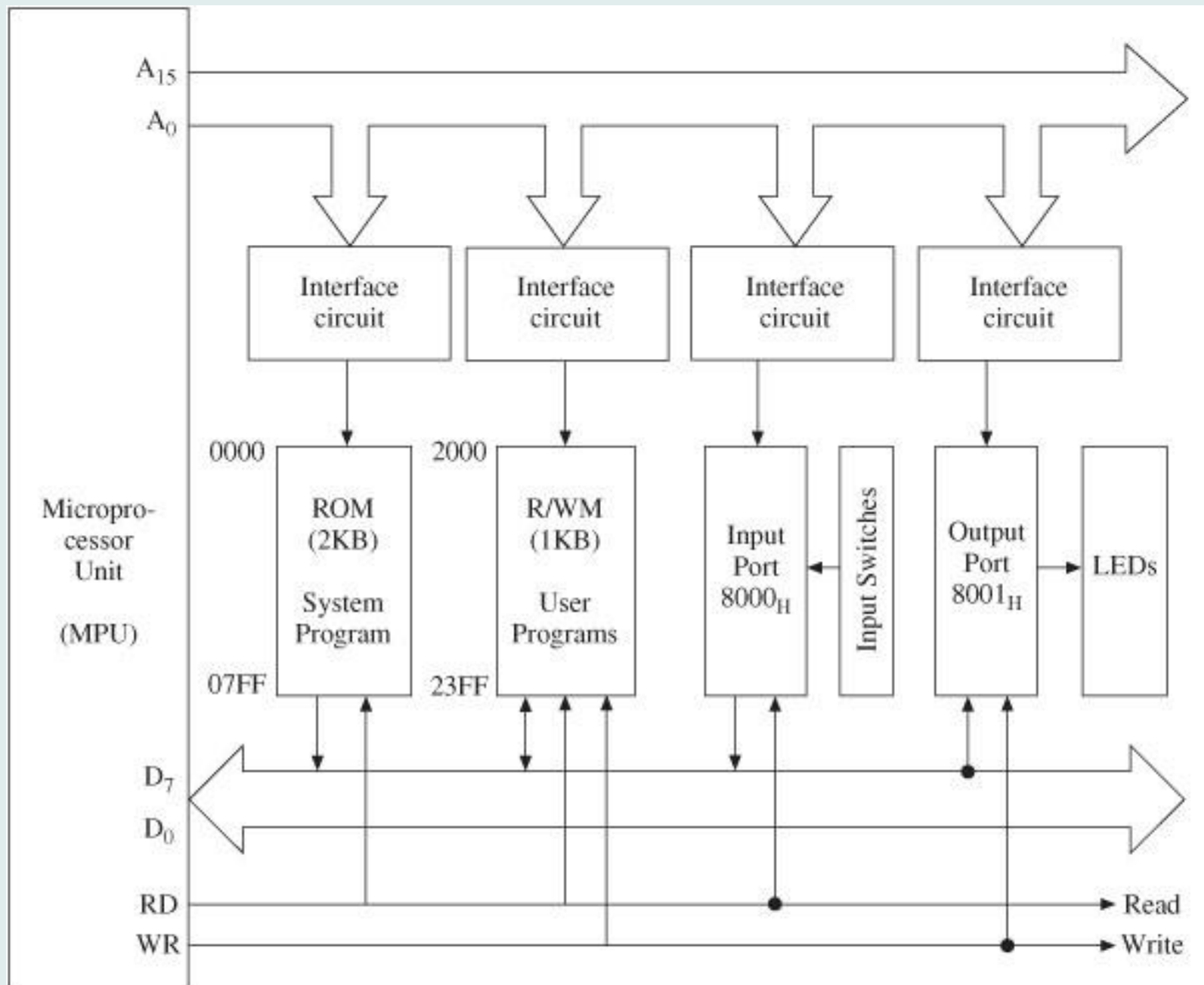
A 16 bit address bus is capable of addressing 2^{16} = 65,536 (64K) addresses.

consists of 16, 20, 24, 32, 64 parallel signal lines

8086 has a 20-bit address bus which corresponds to 2^{20} addresses or 1M (1 Meg) addresses or memory locations;

1





The control bus is used by the microprocessor to send out or receive timing and control signals in order to coordinate and regulate its operation and to communicate with other devices, i.e. memory or input/output.

consists of 4 to 10 (or more) parallel signal lines
CPU sends signals along these lines to memory and to I/O ports

Examples: Memory Read, Memory Write, I/O Read, I/O Write

COMPUTER SOFTWARE

System Software: programs that take control of the PC on start-up, and then play a central role in everything that happens within a computer system by managing, maintaining, and controlling computer resources.

OS, compilers, drivers..

COMPUTER SOFTWARE

Application Software: designed and created to perform specific personal, business, scientific processing tasks.

Word Processors

WordStar

Word perfect

Microsoft Word

Electronic Spreadsheet

Lotus 1-2-3

MS EXCEL

QUATTRO

FRAMEWORK

EMBEDDED SYSTEM

An **embedded system** is a special-purpose computer system designed to perform one or a few dedicated functions often with **real-time**.

An integrated device which consists of multiple devices:

- Microprocessor (MPU)

- Memory

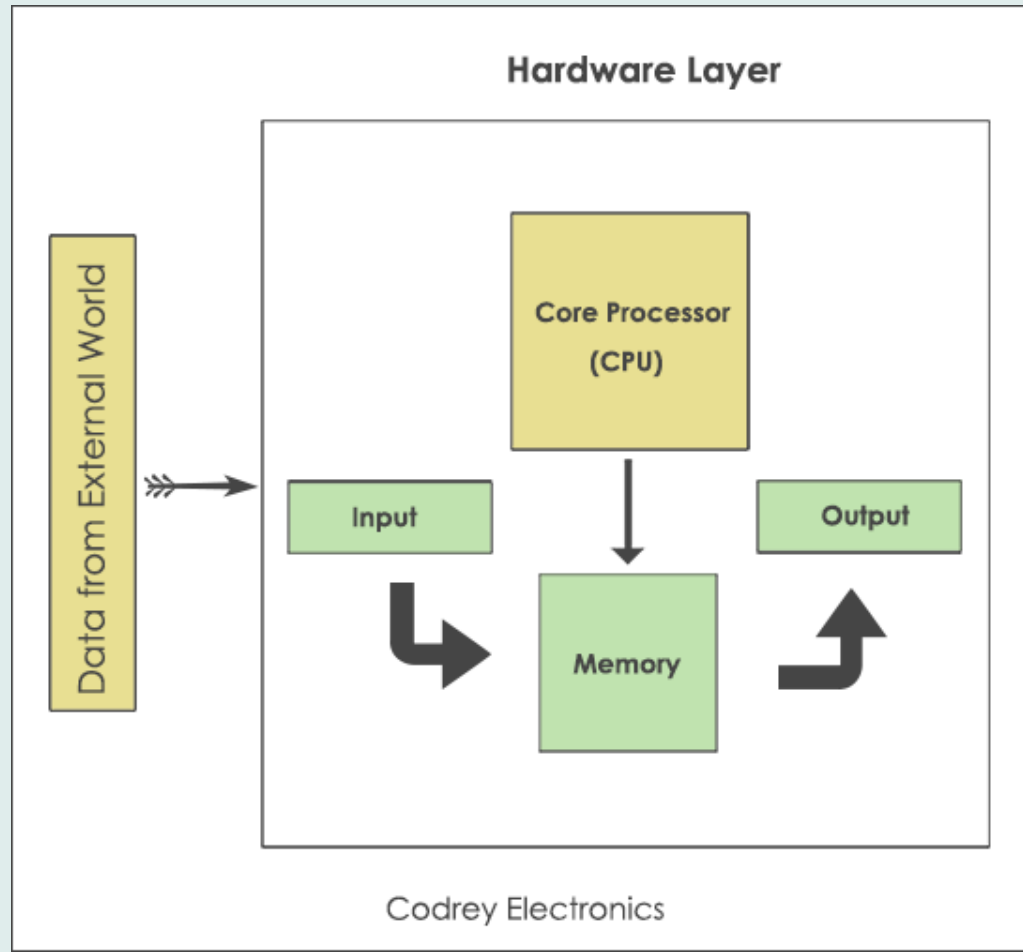
- I/O (Input/Output) ports

Often has its own dedicated software

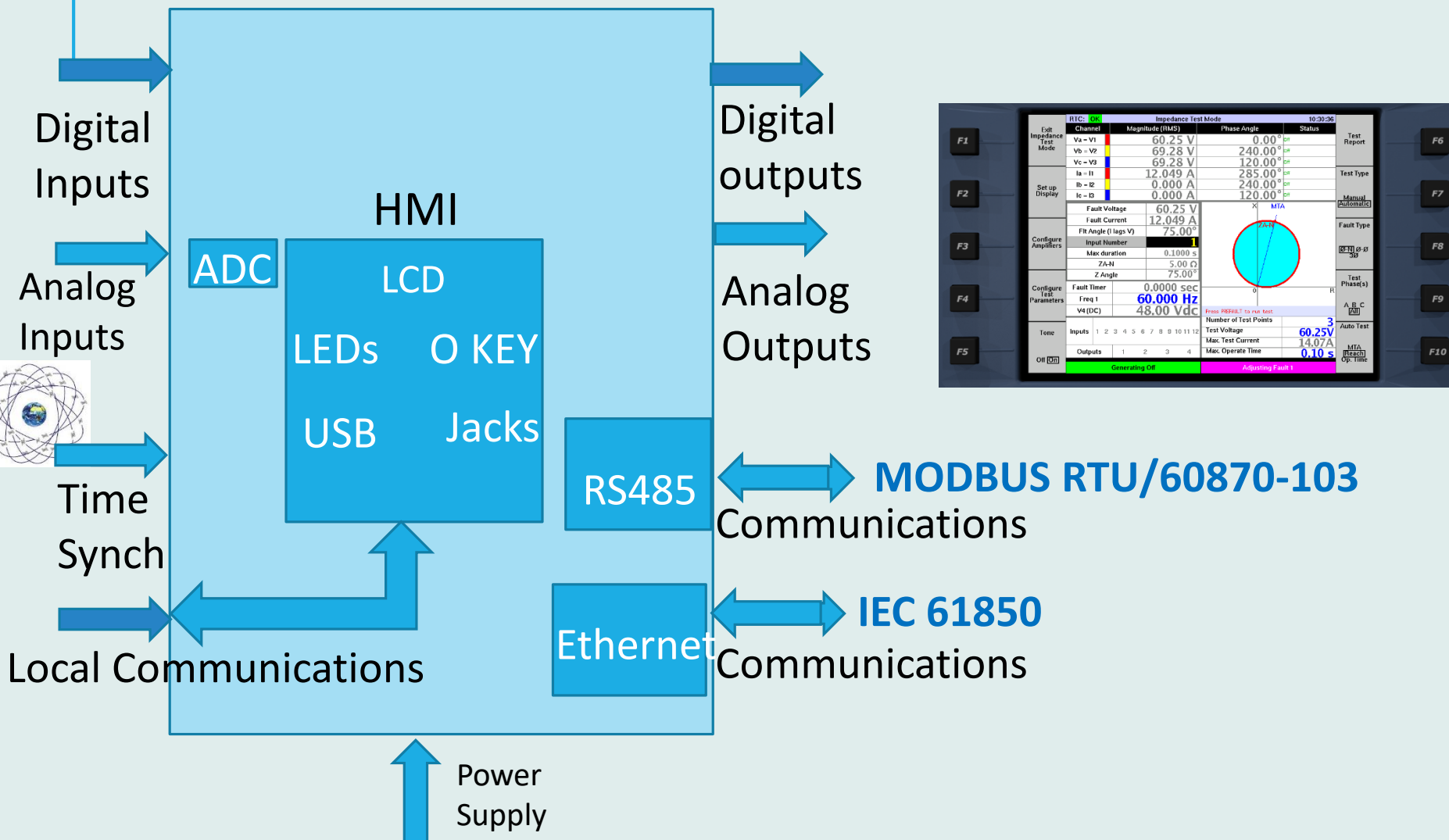
- .System Software

- .Application Software

Example: PS, toys, TV remote, keyless entry, robots, etc
Generally microprocessors or DSP is preferred for embedded system design. Using microcontrollers is cheap but sometimes inadequate for the task.



2.1 The layout of Computer Relays Intelligent Electronic Devices-IEDs



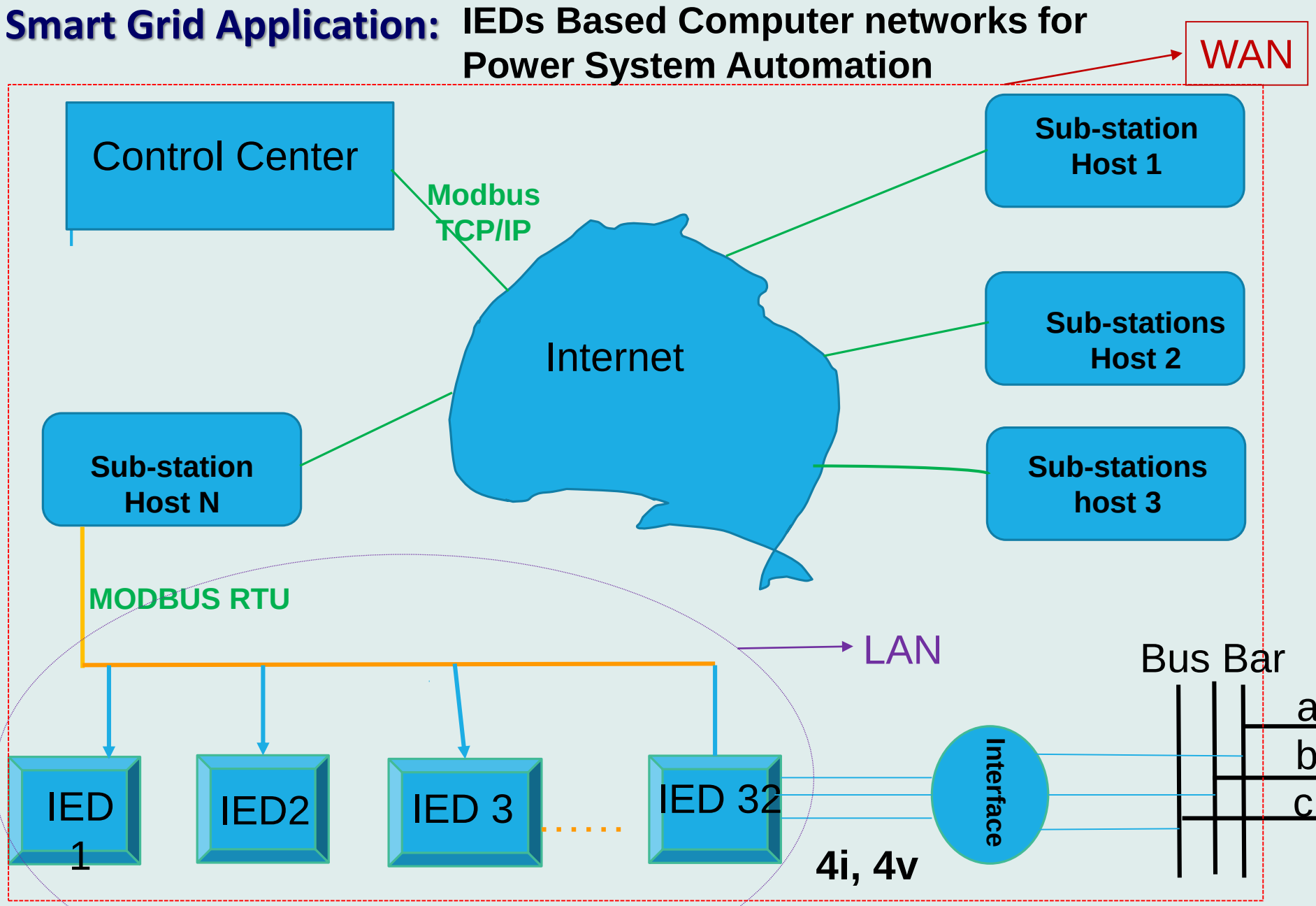
Intelligent Electronic Devices-IEDs

- **Digital inputs:** are used to monitors the states of the power system elements.
- **Digital outputs:** are used to change the states of the power system elements.
- **Analog inputs:** are need to receive voltage and current signal from the power network.
- **Analog outputs:** are need to convey an analogue signal to a display.
- **Time synch:** is need to synchronize the clock of the IED.
- Time synchronizing protocols: SNTP and IRIG-B

- **HMI:** Contains the human machine communication facilities such as operation keys, LCD, LEDs, USB etc.
- **Communication port:** is used to communicate with the master computer within the local operation center, other local and remote IEDs.
- **RS485 port:** is required for communication with Modbus protocol.
- **Ethernet port:** is need to communicate with IEC 61850 protocol. **USB port:** is required to connect a computer to the IED.

- The functions of IEDs can be classified as metering, monitoring, recording, protection, control, managing and communications.
- Some of IEDs could be more advanced than the other. Some of them may emphasize certain functional aspects over the others.
- They could be protective relays, bay controllers, network analyzers, disturbance recorders, RTUs, PMUs etc.
- IEDs gather information of voltages, currents and states of components from electrical power networks via interface unit.
- They use this information and evaluate the operating conditions of power network according to its application program.
- They can also convey these signals to other locally and remotely located IEDs

Smart Grid Application: IEDs Based Computer networks for Power System Automation

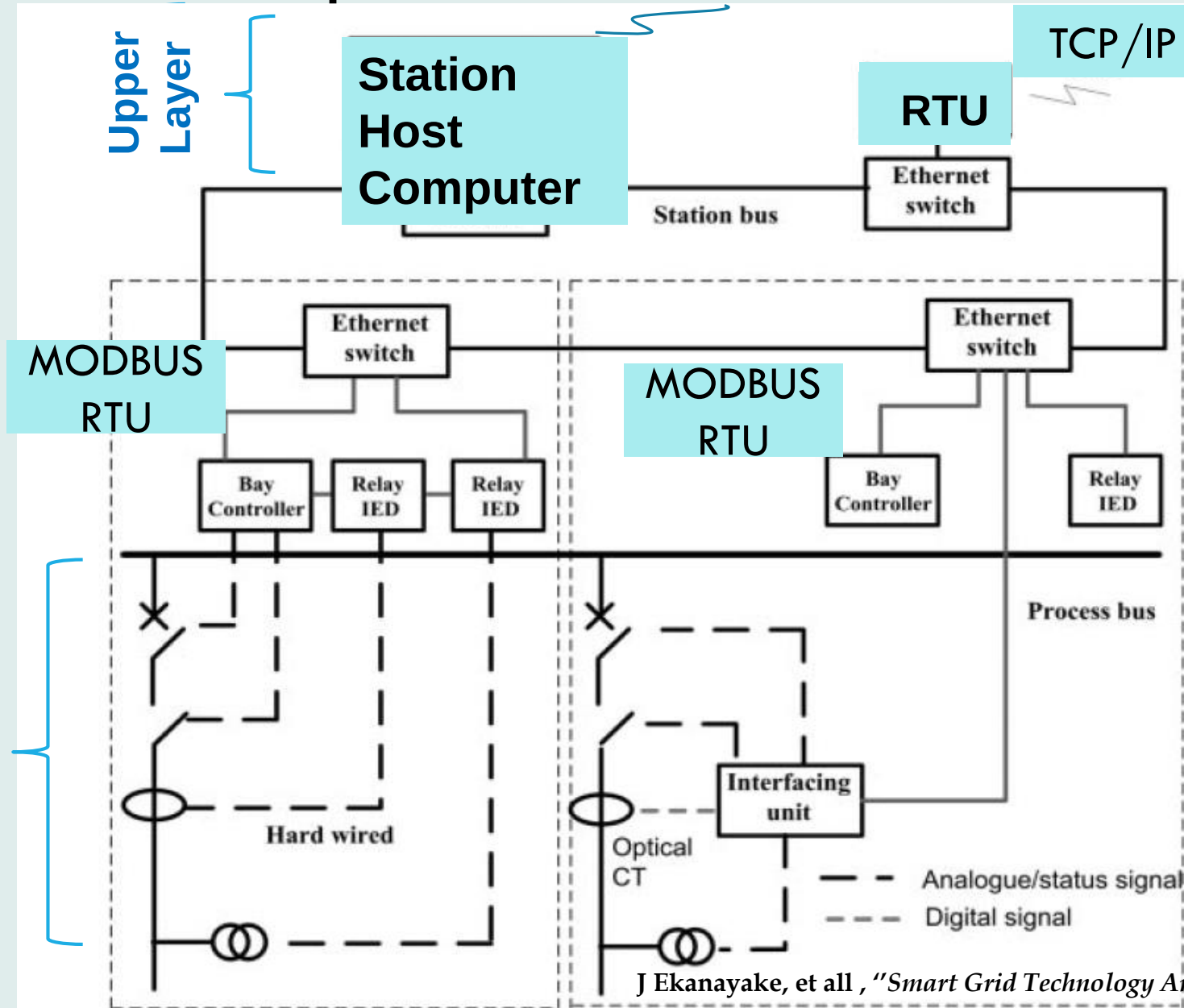


IEDs Based Computer networks for Substation Automation

Lower Layer
Electrical Network

Upper
Layer

Midle Layer



J Ekanayake, et all , "Smart Grid Technology And Applications"

IEDs Based Computer networks for Substation Automation Using IEC 61850 Communications Protocol

Level 3 - Dispatch Center

Control Center

Corporate WAN

Router

Level 2 - Substation SAS

HMI

SCADA

Gateway

61850 Station Bus
Client-Server, GOOSE

Switch

Level 1 - Bays
IEDs

IED

IED

IED

IED

IED

IED

61850 Process Bus
Sampled Values, GOOSE

Switch

Level 0 - Process
Intelligent Switchyard,
Sensors, I/Os

MU

Brkr IED

Brkr IED

MU

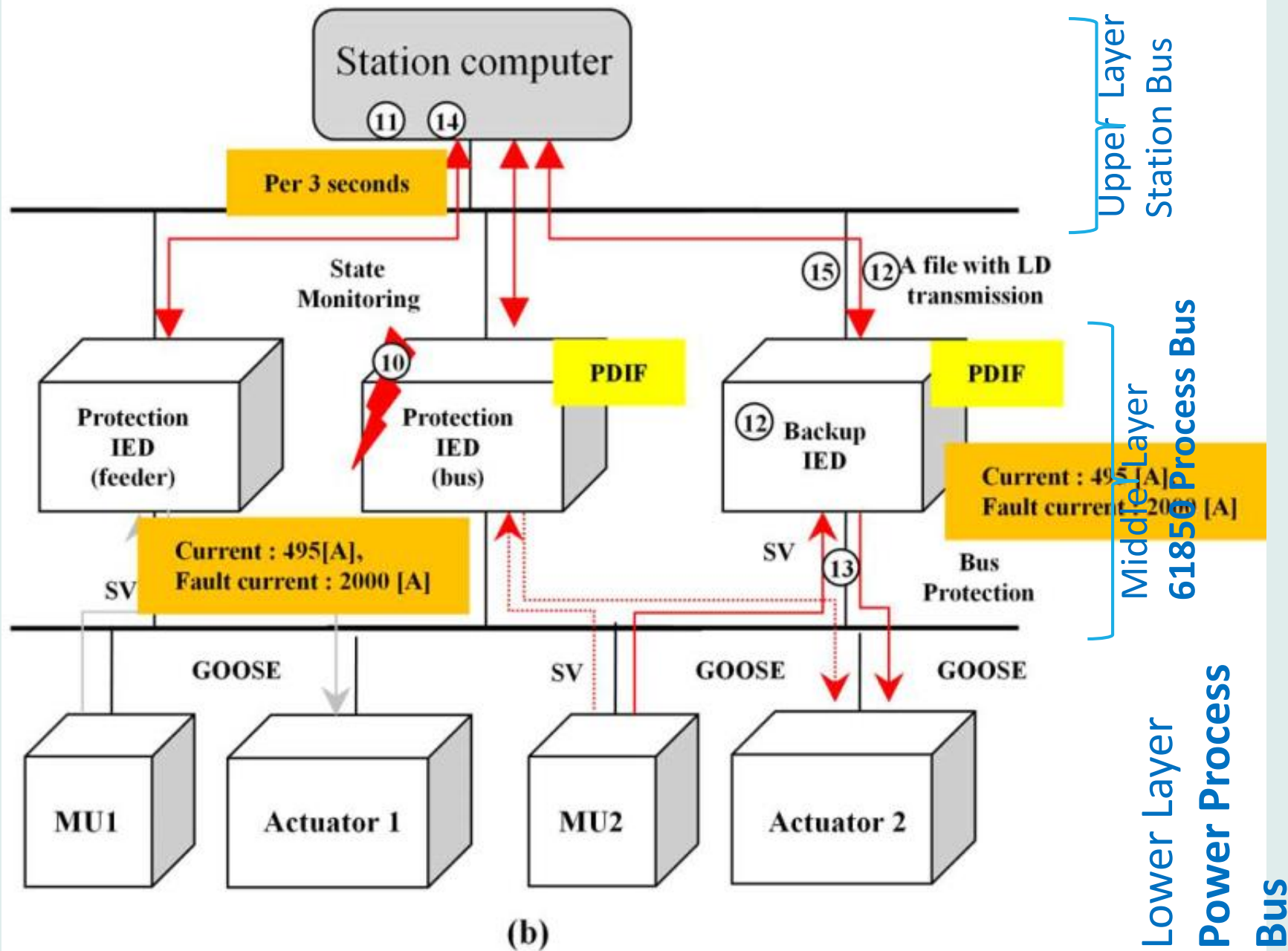
Brkr IED

MU

Upper Layer
Station Bus

Middle Layer
61850
Process Bus

Lower Layer
Power Process
Bus



Intelligent Electronic Devices (IEDs)

I. System Software:

- HMI Software (for USB, LCD, Operation Keys, LEDs)
- Data Acquisition and Collection Software,
- Remote Communications Software (using one or more of following protocols MODBUS RTU, MODBUS TCP/IP, DNP3, IEC60870-103, IEC61850),
- Others.

II. Application (Functions) Software

- Measurement, monitoring, protection, operation, management.